

NPTEL Week 12

ASSIGNMENT 12 SOLUTIONS & EXPLANATIONS

PROGRAMMING SOLUTIONS

1. PageRank Transfer

// PYTHON

```
rank_a, rank_b = [int(x) for x in input().split()]
print(0, rank_a + rank_b, end='')
```

// CONSOLE INJECTION

```
const el = document.querySelector('textarea.inputarea, textarea');
if (el) {
  el.focus();
  document.execCommand('selectAll', false, null);
  document.execCommand('insertText', false, `rank_a, rank_b = [int(x) for x in
input().split()]

print(0, rank_a + rank_b, end='')`);
}
```

2. Mutual PageRank Exchange

// PYTHON

```
rank_a, rank_b = [int(x) for x in input().split()]
print(rank_b, rank_a, end='')
```

// CONSOLE INJECTION

```
const el = document.querySelector('textarea.inputarea, textarea');
if (el) {
  el.focus();
  document.execCommand('selectAll', false, null);
  document.execCommand('insertText', false, `rank_a, rank_b = [int(x) for x in
input().split()]

print(rank_b, rank_a, end='')`);
}
```

3. Three-Page PageRank Update

// PYTHON

```
rank_a, rank_b, rank_c = [int(x) for x in input().split()]
print(0, rank_a // 2, rank_a // 2 + rank_b + rank_c, end='')
```

// CONSOLE INJECTION

```
const el = document.querySelector('textarea.inputarea, textarea');
if (el) {
    el.focus();
    document.execCommand('selectAll', false, null);
    document.execCommand('insertText', false, `rank_a, rank_b, rank_c = [int(x) for x in
input().split()]

print(0, rank_a // 2, rank_a // 2 + rank_b + rank_c, end='')`);
}
```

MCQ & MSQ SOLUTIONS WITH EXPLANATIONS

Scenario 1: Ranking the Most Influential Campus Clubs

Q01 Because it considers the importance of the pages providing the links.

Explanation: PageRank weighs both the quantity and the quality of incoming links. A link from an important page counts for more than a link from an obscure one.

Q02 The new club's PageRank is likely to improve due to the influential incoming link.

Explanation: A highly ranked page passes on a large share of its rank, so the new club gains. It won't automatically jump to the top, though.

Q03 Each outgoing link shares only a smaller portion of the page's ranking value.

Explanation: A page's rank is divided equally among its outgoing links. With hundreds of links, each one carries only a tiny share.

Q04 Importance is determined iteratively based on the network of incoming links.

Explanation: Ranks are recomputed over several rounds from the link structure until they settle. Content length, age and file size play no part.

Q05 Selected Options: **MSQ**

—A link from an influential page generally contributes more than a link from an unimportant page.

—The ranking values are refined over multiple iterations.

Explanation: PageRank depends on link quality, not just the count, and it converges over iterations. Outgoing links pass on rank/out-degree, not zero.

Scenario 2: Ranking Webpages Using a Simplified PageRank Score

Q06 It divides the page's current score equally among its outgoing links.

Explanation: $\text{share} = \text{rank}[\text{page}] / \text{len}(\text{links}[\text{page}])$ splits the page's score evenly among the pages it links to.

Q07 **C**

Explanation: Links: $A \rightarrow [B, C]$, $B \rightarrow [C]$, $C \rightarrow [A, B, D]$, $D \rightarrow [A]$. Totals: $A = 1/3 + 1 = 1.33$, $B = 0.5 + 1/3 = 0.83$, **$C = 0.5 + 1 = 1.5$** , $D = 1/3 = 0.33$.

Q08 **C**

Explanation: C has the most outgoing links (3), so each linked page gets only 1/3. A gives 0.5 per link, and B and D each give 1.

Q09 new_rank

Explanation: Contributions are added into new_rank[dest], and line 17 prints new_rank. rank keeps the original values.

Q10 Selected Options: **MSQ**

—Every page starts with the same initial PageRank value.

—The program performs exactly one PageRank iteration.

Explanation: All pages start at 1.0, and the loop over the pages runs once. More outgoing links means a *smaller* share per link, and rank is never modified.

Scenario 3: Evaluating One Iteration of a Simplified PageRank Algorithm

Q11 R

Explanation: R links only to S, so it passes its entire 1.0 to S. P gives 0.5 per link, Q gives 1/3 and S gives 0.5.

Q12 Q

Explanation: $P \rightarrow [Q, R]$ and $S \rightarrow [P, Q]$. Q is the only page that both P and S link to.

Q13 S

Explanation: $P = 1/3 + 0.5 = 0.83$, $Q = 0.5 + 0.5 = 1.0$, $R = 0.5 + 1/3 = 0.83$, **$S = 1/3 + 1 = 1.33$** .

Q14 To ensure every contribution is computed using the original ranks.

Explanation: If rank were updated mid-loop, pages processed later would distribute their new values instead of their original ones, which would give wrong results.

Q15 Selected Options: **MSQ**

—A page with fewer outgoing links transfers a larger share to each linked page.

—The total contribution distributed by each page is equal to its current rank.

—All updated ranks are computed before replacing the original ranks.

Explanation: Share = rank/out-degree, so fewer links means a bigger share, and the shares add back up to the page's rank. All values go into updated while rank stays unchanged during the loop.

Scenario 4: Exploring the Collatz Conjecture

Q16 3

Explanation: $8 \rightarrow 4 \rightarrow 2 \rightarrow 1$. The loop runs 3 times, so steps += 1 (line 10) executes 3 times.

Q17 It checks whether the current number is even.

Explanation: if $n \% 2 == 0$: is true when n is divisible by 2.

Q18 5

Explanation: Input 3 prints 3, then becomes 10. 10 is printed, and because it is even it is halved to 5. 16 would come from wrongly applying $3n+1$ to 10.

Q19 It counts the number of transformations performed before reaching 1.

Explanation: steps increases once per loop iteration, and each iteration applies one transformation.

Q20 Selected Options: **MSQ**

- The value 1 is printed even though the loop terminates before processing it.
- The variable steps is incremented exactly once during every loop iteration.
- The program prints every intermediate value generated by the sequence.

Explanation: while $n \neq 1$ stops before printing 1, so print(1) on line 12 prints it. For odd n , $3n+1$ is always *larger*, so that option is false.
